

# CarbonSense Policy Intelligence & Compliance



Full Implementation Plan

India-Focused · SME-First · LLM-Assisted · Data-Verified

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## 1. Project Overview & Objectives

CarbonSense is a carbon intelligence platform for SMEs and enterprises, combining emissions operations, AI-assisted analytics, OCR receipt processing, and the TEME (Tree-Emission Matching Engine). The Policy Intelligence and Compliance modules are currently implemented as static mock screens. This document provides a complete plan to transform them into a fully data-driven, LLM-assisted, India-specific compliance management system.

### Core Design Principle

The power of the system is NOT in the number of policies listed. It is in how well those policies are translated into **measurable, verifiable checks**. Every compliance requirement must be provable through data change, evidence upload, or OCR extraction — never by a user simply clicking “mark complete”.

### Primary Target

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| Primary audience   | Small and Medium Enterprises (SMEs) across India                                |
| Secondary audience | Larger enterprises and listed companies                                         |
| Geographic focus   | India — regulatory and policy framework                                         |
| Guiding philosophy | Reduction-first. Offsets as projected mitigation, not immediate neutralisation. |
| Compliance model   | Data + Actions + Verification (not self-declaration)                            |

2. Current State Assessment

What Is Already Built

- Next.js frontend with persistent sidebar, dark theme, teal/emerald design system
- FastAPI backend with ingestion, OCR, recommendations, and TEME endpoints
- Supabase persistence for emissions uploads, recommendation sessions, and TEME runs
- Policy Intelligence page: static UI with CCUS hero, compliance cards, funding grid, completed actions
- Compliance page: static KPI cards, task list with progress bars, deadline buckets
- Emissions ingestion engine (CSV + manual) with Scope 1/2/3 calculations
- OCR receipt processing pipeline — single and bulk
- Recommendation generation with LLM + heuristic fallback

Key Gaps to Address

| Area               | Current State                | Target State                        |
|--------------------|------------------------------|-------------------------------------|
| Policy data        | Hardcoded arrays in page.tsx | Seeded Supabase table, API-driven   |
| Compliance data    | Static mock objects          | Per-org results in Supabase         |
| Score calculation  | Not implemented              | Weighted formula (40/40/20)         |
| Verification       | User self-declaration        | Data change + evidence upload + OCR |
| LLM on policy page | Not present                  | Groq-powered policy explainer chat  |
| LLM on compliance  | Not present                  | Score improvement suggestions       |
| Evidence upload    | Not wired                    | OCR pipeline → compliance_results   |
| Score history      | Not present                  | Time-series snapshots in Supabase   |

### 3. Indian Policy Ecosystem — Complete Catalogue

India's environmental policy landscape spans seven layers. The table below lists every policy relevant to CarbonSense, with implementation priority for the current build.

| Policy / Scheme                                 | Authority                          | Layer             | Priority   | Applicability              |
|-------------------------------------------------|------------------------------------|-------------------|------------|----------------------------|
| Carbon Credit Trading Scheme (CCTS)             | MoEFCC / BEE                       | Carbon market     | Core       | Large industry, future SME |
| National Action Plan on Climate Change (NAPCC)  | PMO / MoEFCC                       | Climate           | Background | All                        |
| PAT Scheme (Perform, Achieve, Trade)            | Bureau of Energy Efficiency        | Energy efficiency | Core       | Manufacturing SMEs         |
| Energy Conservation Act 2001 (amended 2022)     | BEE                                | Energy            | Core       | All                        |
| UJALA / LED Programme                           | EESL                               | Energy            | Optional   | All                        |
| Solar Subsidy — Rooftop (MNRE)                  | Ministry of New & Renewable Energy | Renewable         | Core       | All SMEs                   |
| Renewable Purchase Obligation (RPO)             | MNRE / SERCs                       | Renewable         | Secondary  | Large consumers            |
| Renewable Energy Certificates (REC)             | CERC                               | Renewable         | Advanced   | Large enterprises          |
| BRSR (Business Responsibility & Sustainability) | SEBI                               | ESG reporting     | Core       | Listed cos; indirect SMEs  |
| ESG Framework (GRI / GHG Protocol)              | International                      | ESG               | Core       | All                        |
| Environmental Protection Act (EPA) 1986         | MoEFCC                             | Environmental     | Core       | All                        |
| Air (Prevention & Control of Pollution) Act     | CPCB / SPCBs                       | Environmental     | Core       | Manufacturing              |
| Water (Prevention & Control of Pollution) Act   | CPCB                               | Environmental     | Secondary  | Manufacturing              |
| Hazardous Waste Management Rules 2016           | MoEFCC                             | Waste             | Core       | Factories                  |

|                                              |                              |               |           |                                |
|----------------------------------------------|------------------------------|---------------|-----------|--------------------------------|
| Solid Waste Management Rules 2016            | MoEFCC                       | Waste         | Core      | All                            |
| Plastic Waste Management Rules 2022          | MoEFCC                       | Waste         | Core      | Restaurant, retail             |
| E-Waste Management Rules 2022                | MoEFCC                       | Waste         | Optional  | IT sector                      |
| FAME Scheme (EV Policy)                      | Ministry of Heavy Industries | Transport     | Core      | Logistics SMEs                 |
| Bharat Stage Emission Norms (BS-VI)          | MoRTH                        | Transport     | Secondary | Fleet operators                |
| ZED Certification (Zero Defect Zero Effect)  | Quality Council of India     | MSME          | Core      | Manufacturing SMEs             |
| MSME Sustainable Finance Schemes             | SIDBI / Ministry of MSME     | MSME          | Core      | All SMEs                       |
| Credit Linked Capital Subsidy Scheme (CLCSS) | Ministry of MSME             | MSME          | Core      | Manufacturing SMEs             |
| National Green Hydrogen Mission              | MNRE                         | Energy        | Future    | Manufacturing, large SMEs      |
| Extended Producer Responsibility (EPR)       | MoEFCC                       | Waste         | Core      | Packaged goods, electronics    |
| MSME Green Rating Scheme                     | BIS / MoMSME                 | MSME          | Core      | All SMEs                       |
| ISO 14001 Alignment                          | ISO (via BIS in India)       | Environmental | Secondary | Export-oriented SMEs           |
| GHG Protocol Alignment                       | WRI / WBCSD                  | Reporting     | Core      | All (engine already uses this) |
| Carbon Border Adjustment Mechanism (CBAM)    | European Union               | Trade         | Future    | EU-exporting SMEs              |
| BRSR Core (mandatory top 150 listed cos)     | SEBI                         | ESG           | Secondary | Listed companies               |

## 4. Compliance Framework — Six Categories

In the CarbonSense system, compliance is modelled as three dimensions: Data + Actions + Verification. These are grouped across six regulatory categories, each broken into three implementation levels.

### A — Emission & Energy Compliance

- Energy Consumption Tracking — measure electricity usage (kWh)
- Fuel Consumption Tracking — diesel, petrol, LPG in litres
- Emission Calculation — Scope 1, 2, 3 per GHG Protocol (engine already does this)
- Energy Efficiency Improvement — reduce energy intensity per unit output (PAT Scheme)

### B — Waste Management Compliance

- Waste Quantity Tracking — kg/month by waste stream
- Waste Segregation — wet / dry / hazardous separation documented
- Proper Disposal — authorised vendor invoices uploaded
- Plastic Waste Compliance — single-use reduction metrics
- Hazardous Waste Handling — safe storage certificates and disposal records

### C — Environmental Regulatory Compliance

- Pollution Control Board (PCB) Registration — certificate upload
- Air Emission Standards — compliance with CPCB limits
- Water Pollution Compliance — wastewater treatment records (manufacturing)
- Environmental Clearance — where applicable by industry scale

### D — ESG & Reporting Compliance

- Emission Reporting — annual or periodic submission
- ESG Data Collection — energy, waste, emissions in structured format
- BRSR Reporting — required for listed companies; preparedness for SMEs
- GHG Protocol alignment — already implemented in emission engine

### E — Carbon & Market Compliance

- Carbon Intensity Tracking — CO2 per unit of output
- Carbon Credit Compliance (CCTS) — meet intensity targets or procure credits
- TEME integration — tree-based offset planning with projected, not verified, language

F — Operational & Action Compliance (CarbonSense Innovation)

- Policy Action Tracking — steps completed per applied recommendation
- Evidence Submission — proof of action (invoices, certificates, photos)
- Verification — data-change or document-based, never self-declaration

Three Implementation Levels

| Level                       | Target           | Requirements                                                                                                                        |
|-----------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Level 1 — Basic             | ALL SMEs         | Energy tracked · Fuel tracked · Emissions calculated · Waste tracked · Waste segregated · ESG data basic                            |
| Level 2 — Industry-specific | Sector-dependent | Manufacturing: fuel, hazardous waste, pollution norms Restaurant: food waste, LPG, plastic Logistics: fuel usage, vehicle emissions |
| Level 3 — Action-based      | Policy adopters  | Policy applied · Steps progressed · Completion verified · Evidence uploaded                                                         |

5. Compliance Score Model

The compliance score is a weighted composite of three dimensions:

$$\text{total\_score} = \text{data\_score} + \text{action\_score} + \text{reporting\_score}$$

| Dimension            | Weight | Formula                                                    | Max |
|----------------------|--------|------------------------------------------------------------|-----|
| Data compliance      | 40%    | completed data requirements / total data requirements × 40 | 40  |
| Action compliance    | 40%    | completed + verified action requirements / total × 40      | 40  |
| Reporting compliance | 20%    | submitted reports / expected reports × 20                  | 20  |
| TOTAL                | 100%   | Sum of above three dimensions                              | 100 |

Verification Rules — What Counts as Complete

A requirement only increments the score if the following conditions are met. User self-declaration is **never accepted**.

| Requirement type | Verification condition                                                                                                                                    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data type        | The relevant metric (electricity kWh, fuel litres, waste kg) exists in org emissions data AND shows a measurable change from the previous period baseline |
| Action type      | Evidence document has been uploaded AND optionally extracted via OCR to confirm the relevant metric or certificate number                                 |
| Reporting type   | A reporting artifact has been submitted with a recorded timestamp and linked to the correct period                                                        |

## 6. Supabase Database Schema

Five new tables are required. All existing emissions and recommendation tables remain unchanged.

### policies — Master Policy Catalogue

| Column         | Type       | Description                                                                         |
|----------------|------------|-------------------------------------------------------------------------------------|
| id             | uuid PK    | Primary key                                                                         |
| name           | text       | Full policy name e.g. 'PAT Scheme'                                                  |
| short_name     | text       | Abbreviation e.g. 'PAT'                                                             |
| category       | text       | energy / renewable / waste / environmental / esg / carbon_market / transport / msme |
| layer          | text       | core / secondary / optional / future                                                |
| description    | text       | Plain-language summary for display                                                  |
| authority      | text       | Governing body e.g. 'Bureau of Energy Efficiency'                                   |
| applicability  | text[]     | sme / large_enterprise / manufacturing / restaurant / logistics / retail            |
| requirements   | text[]     | What the policy mandates                                                            |
| benefits       | text[]     | Financial, legal, reputational benefits                                             |
| is_active      | boolean    | Whether policy is current                                                           |
| effective_date | date       | When policy came into force                                                         |
| review_date    | date       | Next review or expiry date                                                          |
| external_url   | text       | Link to official government source                                                  |
| created_at     | timestampz | Record creation timestamp                                                           |

## compliance\_requirements — Individual Checks

| Column              | Type    | Description                                          |
|---------------------|---------|------------------------------------------------------|
| id                  | uuid PK | Primary key                                          |
| policy_id           | uuid FK | References policies.id                               |
| name                | text    | e.g. 'Electricity consumption tracked'               |
| type                | text    | data / action / reporting                            |
| level               | text    | basic / industry_specific / action_based             |
| industry            | text[]  | Applicable industries                                |
| verification_method | text    | data_change / evidence_upload / ocr_extract / manual |
| description         | text    | What the user must do                                |
| is_mandatory        | boolean | Mandatory vs recommended                             |
| weight              | numeric | Contribution to score calculation                    |

## compliance\_results — Per-Org Status

| Column              | Type       | Description                                                |
|---------------------|------------|------------------------------------------------------------|
| id                  | uuid PK    | Primary key                                                |
| organization_id     | uuid       | Organisation reference                                     |
| requirement_id      | uuid FK    | References compliance_requirements.id                      |
| status              | text       | not_started / in_progress / completed / overdue / verified |
| verified            | boolean    | True only after data or evidence confirmed                 |
| verification_source | text       | data / evidence / ocr / llm                                |
| evidence_url        | text       | URL of uploaded proof document                             |
| data_snapshot       | jsonb      | Captured metric value at time of verification              |
| score_contribution  | numeric    | Points this result contributes to total score              |
| due_date            | date       | Deadline for this requirement                              |
| completed_at        | timestampz | When verified status was set                               |
| notes               | text       | Free-text notes or LLM extraction summary                  |
| updated_at          | timestampz | Last update timestamp                                      |

## compliance\_score\_history — Time-Series Scores

| Column          | Type    | Description                                |
|-----------------|---------|--------------------------------------------|
| id              | uuid PK | Primary key                                |
| organization_id | uuid    | Organisation reference                     |
| snapshot_date   | date    | Date of this score snapshot                |
| data_score      | numeric | Data dimension score (max 40)              |
| action_score    | numeric | Action dimension score (max 40)            |
| reporting_score | numeric | Reporting dimension score (max 20)         |
| total_score     | numeric | Total score out of 100                     |
| breakdown       | jsonb   | Per-category breakdown for chart rendering |

## policy\_interactions — LLM Conversation Log

| Column          | Type       | Description                                             |
|-----------------|------------|---------------------------------------------------------|
| id              | uuid PK    | Primary key                                             |
| organization_id | uuid       | Organisation reference                                  |
| policy_id       | uuid FK    | References policies.id (nullable for general questions) |
| user_question   | text       | What the user asked                                     |
| llm_response    | text       | Response from Groq LLM                                  |
| model_used      | text       | Model identifier e.g. 'llama-3.1-8b-instant'            |
| created_at      | timestampz | Timestamp of interaction                                |

## 7. Policy Intelligence Page — Redesign Plan

The Policy Intelligence page transitions from a single static CCUS banner with hardcoded cards to a fully data-driven policy management experience. The dark UI, teal/emerald accents, and card layout are preserved.

### Section 1 — Policy Radar Grid

- Replaces the static CCUS hero banner
- Grid of policy cards filtered by the org's industry profile (stored in org settings)
- Each card shows: policy name, authority, category badge, applicability tags, status pill
- Status pill values: Applicable · Not Applicable · Future Readiness · In Progress · Compliant
- Filter controls: by category (Energy / Waste / ESG / Carbon / MSME / Transport)
- Search by policy name for quick access
- Clicking any card opens a detail drawer (Section 2)

### Section 2 — Policy Detail Drawer

- Slides in from the right, preserving grid context
- Shows: full policy description, authority, effective date, official source link
- Requirements list — what this policy mandates for the org
- Benefits list — financial, legal, and reputational benefits of compliance
- Current compliance status for this policy (progress bar across requirements)
- LLM Chat Panel — user asks questions, Groq responds in context
- Example questions surfaced as suggestions: 'Does this apply to us?', 'What should we do first?', 'What documents do we need?'
- All interactions logged to policy\_interactions table

### Section 3 — Funding & Incentives Grid

- Retained from current design but made data-driven
- Sourced from policies table where benefits contains financial keywords
- Cards show: scheme name, amount/benefit, eligibility, deadline, apply CTA
- Key schemes: CLCSS, ZED certification subsidy, MNRE solar subsidy, MSME sustainable finance
- New addition: FAME EV subsidy for logistics SMEs

### Section 4 — Policy Action Tracker

- Bridge between Policy and Compliance pages
- For each applicable policy, shows compliance\_requirements mapped to it
- Each requirement row shows: name, type badge, verification method, current status
- Quick action button routes to Compliance page with that requirement pre-highlighted
- Provides policy-centric view vs Compliance page's task-centric view

## Section 5 — Completed Policy Actions (Audit Trail)

- Retained from current design, made data-driven from compliance\_results
- Filters to verified = true with evidence\_url present
- Shows: requirement name, policy, completion date, verification source badge
- Exportable as PDF audit trail

## 8. Compliance Page — Redesign Plan

The Compliance page transitions from static KPI cards and mock task lists to a live compliance management dashboard backed by real Supabase data. The emerald/amber/rose status colour system and progress-centric design are preserved and extended.

### Section 1 — Score Dashboard

- Three animated score rings: Data (max 40) · Action (max 40) · Reporting (max 20)
- Total score prominently displayed with colour coding: ≥80 green · 50–79 amber · <50 red
- Sparkline chart below rings showing score trend from `compliance_score_history`
- Score breakdown tooltip showing per-category contribution
- LLM-generated one-line improvement tip (Groq call on page load, cached for 24h)

### Section 2 — Compliance by Level (Three Tabs)

- Tab 1 — Basic (all SMEs): energy, fuel, emission, waste, segregation, ESG data
- Tab 2 — Industry-specific: manufacturing / restaurant / logistics requirements
- Tab 3 — Action-based: requirements linked to applied policies and recommendations
- Each tab shows requirements as rows with: name, type badge, verification method, status pill
- Status colours: `not_started` grey · `in_progress` amber · `completed` green · `overdue` red · `verified` teal

### Section 3 — Active Tasks

- Pulled from `compliance_results` where status is `not_started` or `in_progress`
- Each task card shows: requirement name, parent policy, type badge, verification method, days to due date
- Context-aware action button:
- Data type → 'View your emissions data' (links to Detailed Log filtered to relevant metric)
- Action type → 'Upload evidence' (opens evidence upload panel)
- Reporting type → 'Submit report' (opens reporting flow)
- Priority sorting: overdue first, then by days remaining ascending

### Section 4 — Evidence Upload Panel

- Triggered from 'Upload evidence' on any action-type requirement
- Accepts: PDF, JPG, PNG (invoices, certificates, utility bills, vendor receipts)
- Routes through existing OCR pipeline
- OCR extracts: relevant metric value, date, vendor/authority name, certificate number
- Extraction result shown to user for confirmation before saving
- On confirm: updates `compliance_results` with `verified=true`, `evidence_url`, `data_snapshot`, `verification_source='ocr'`
- LLM (Groq) reviews OCR output and flags if extracted data does not match requirement

## Section 5 — Completed & Verified (Audit Trail)

- Data-driven from compliance\_results where verified=true
- Shows: requirement name, policy, verification source badge, completion date
- Evidence link opens uploaded document
- Exportable as signed audit trail PDF

## Section 6 — Upcoming Deadlines (Next 90 Days)

- Data-driven from due\_date on compliance\_results
- Grouped by month bucket (same as current design but real data)
- Status badges: on track / at risk / overdue
- Days remaining chip colour-coded: >30 green · 10–30 amber · <10 red

9. LLM Integration — Groq vs OpenRouter

Recommendation: Use Groq

Groq is the correct choice for the Policy and Compliance LLM features. Here is the full comparison against OpenRouter:

| Factor            | Groq                                                   | OpenRouter (free tier)                         |
|-------------------|--------------------------------------------------------|------------------------------------------------|
| Speed             | Fastest available (LPU hardware, ~500 tok/s)           | Slower, routes through community models        |
| Free tier         | Generous — 14,400 req/day on llama-3.1-8b-instant      | Free models vary, inconsistent availability    |
| API compatibility | OpenAI-compatible (fits your existing abstraction)     | OpenAI-compatible                              |
| Model quality     | llama-3.1-8b-instant — excellent for explanation tasks | Model varies, less predictable quality         |
| Latency UX        | Chat panel feels responsive (<1s typical)              | Higher latency breaks chat feel                |
| Cost at scale     | Very competitive paid tier                             | Aggregator markup                              |
| Verdict           | USE THIS for policy & compliance chat                  | Keep as fallback for recommendation generation |

Environment Configuration

Set these in your backend .env to activate Groq:

```
LLM_PROVIDER=groq
LLM_API_BASE_URL=https://api.groq.com/openai/v1
LLM_MODEL=llama-3.1-8b-instant
LLM_API_KEY=your_groq_api_key_here
```

Your existing provider abstraction already supports OpenAI-compatible endpoints. No code changes are needed to the recommendation service — just update the env vars.

LLM Roles — What It Does and Does NOT Do

On the Policy Intelligence page:

- Policy explainer — given the policy details + org profile, explains in plain language
- Applicability advisor — answers 'does this apply to our business?'
- Action suggester — 'what should we do first to comply with PAT Scheme?'
- Document interpreter — explains compliance certificates or regulatory notices

On the Compliance page:

- Score improvement — 'your data score is low because electricity tracking is missing, here is how to fix it'

- OCR verification assist — reads extracted text and confirms if it satisfies the requirement
- Gap analysis — identifies which requirements are closest to completion
- Plain-language compliance report generation

**What LLM must NEVER do:**

- Set verified = true directly — only the rule engine does this after data or evidence confirmation
- Override the score calculation — scores come from the database, not from LLM output
- Make legal determinations — it explains, suggests, and extracts; it does not certify compliance

**System Prompt Template for Policy Chat**

Pass this context to Groq on every policy question:

```
You are a carbon compliance advisor for Indian SMEs.  
Organisation industry: {industry}  
Organisation size: {size}  
Current emissions: {total_emissions} kgCO2e  
Policy context: {policy_name} – {policy_description}  
Policy requirements: {requirements_list}  
Answer in plain language. Be specific and actionable.  
Do not make legal determinations. Recommend consulting a CA or legal advisor for binding advice.
```

## 10. Backend API Endpoints

Add these FastAPI endpoints to the existing router group structure.

| Method | Endpoint                        | Description                                                                   |
|--------|---------------------------------|-------------------------------------------------------------------------------|
| GET    | /compliance/requirements        | List requirements filtered by org industry and level                          |
| GET    | /compliance/results             | Get org's compliance results with status and scores                           |
| POST   | /compliance/results/{id}/verify | Trigger verification — checks data or saves evidence                          |
| POST   | /compliance/score               | Recalculate and persist score snapshot for org                                |
| GET    | /compliance/score/history       | Return time-series score data for trend chart                                 |
| GET    | /policies                       | List all active policies with optional category and industry filters          |
| GET    | /policies/{id}                  | Full policy detail including requirements and benefits                        |
| POST   | /policies/ask                   | LLM chat — takes policy_id, org context, user question; returns Groq response |
| GET    | /compliance/deadlines           | Upcoming deadlines for org in next 90 days                                    |
| POST   | /compliance/evidence            | Upload evidence document, run OCR, return extraction for confirmation         |

Mount these under a new /compliance and /policies router group in packages/ml\_services/api/app.py, consistent with the existing /teme, /ocr, /recommendations, and /ingestion groups.

## 11. Implementation Sequence

Follow this order strictly. Building UI on unstable data models creates expensive rework. Each phase has clear deliverables before the next phase starts.

### Phase 1 — Data Foundation

- Write and run Supabase migration for all five new tables
- Seed policies table with the full 29-policy catalogue from Section 3
- Seed compliance\_requirements with all requirements mapped to policies
- Verify foreign key relationships and RLS policies
- Deliverable: populated database ready for API layer

### Phase 2 — Backend API Layer

- Create /compliance and /policies FastAPI router modules
- Implement all 10 endpoints from Section 10
- Add Groq client to existing LLM provider abstraction
- Wire /policies/ask to Groq with org-context system prompt
- Wire /compliance/evidence through existing OCR pipeline
- Add score calculation logic (40/40/20 formula)
- Deliverable: all endpoints tested via FastAPI /docs

### Phase 3 — Compliance Page

- Replace static KPI cards with live score rings from /compliance/score
- Replace static task list with /compliance/results data
- Implement three-tab level view (Basic / Industry / Action)
- Build evidence upload panel wired to /compliance/evidence
- Add score history sparkline from /compliance/score/history
- Add LLM improvement tip on page load (cached)
- Deliverable: fully data-driven compliance page

### Phase 4 — Policy Intelligence Page

- Replace static cards with policy grid from /policies
- Add industry and category filter controls
- Build policy detail drawer with full information
- Integrate LLM chat panel using /policies/ask
- Wire Funding section to policies with financial benefits
- Build Policy Action Tracker section
- Deliverable: fully data-driven policy page with LLM chat

## Phase 5 — Automation & Polish

- Add score snapshot trigger — record to `compliance_score_history` on each verification
- Add due date notification system — surface overdue items in dashboard
- Wire policy compliance status into executive dashboard policy snapshot widget
- Add export to PDF for audit trail (policy interactions + compliance results)
- Performance testing on Groq chat latency
- Deliverable: production-ready compliance management system

## 12. Missing Policies — Additions to Current List

The following policies were missing from the original framework document and should be added to the policies table. These are significant for SME completeness.

### National Green Hydrogen Mission

|                    |                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | MNRE — 2023                                                                                                                                          |
| Why it matters     | Core for large manufacturing SMEs. Green hydrogen as alternative fuel. Relevant for energy-intensive industries looking to decarbonise process heat. |
| Applies to         | Manufacturing, large SMEs                                                                                                                            |

### Extended Producer Responsibility (EPR)

|                    |                                                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | MoEFCC — Plastic Waste Rules 2022                                                                                                                    |
| Why it matters     | Mandatory for ANY SME that manufactures packaged goods, electronics, or uses plastic packaging. Requires take-back programmes and reporting to CPCB. |
| Applies to         | All packaged goods manufacturers, electronics                                                                                                        |

### MSME Green Rating Scheme

|                    |                                                                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | BIS / Ministry of MSME                                                                                                                                                |
| Why it matters     | Government sustainability rating that provides access to green financing, preferential procurement, and export credentials. Highly relevant for export-oriented SMEs. |
| Applies to         | All SMEs                                                                                                                                                              |

### ISO 14001 Environmental Management

|                    |                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | ISO (via BIS in India)                                                                                                                        |
| Why it matters     | Not a law but practically mandatory for export-oriented SMEs and often a procurement prerequisite. Shows systematic environmental management. |
| Applies to         | Export-oriented SMEs                                                                                                                          |

### GHG Protocol Alignment (formalised)

|                    |                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | WRI / WBCSD                                                                                                                              |
| Why it matters     | Your engine already uses this. Surface it explicitly as a compliance item to add credibility and align with BRSR reporting requirements. |
| Applies to         | All (engine already implements this)                                                                                                     |

## CBAM — Carbon Border Adjustment Mechanism

|                    |                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | European Union (external)                                                                                                                         |
| Why it matters     | Future-readiness item for any SME exporting to Europe. EU will charge carbon tariffs from 2026 on products from countries without carbon pricing. |
| Applies to         | EU-exporting SMEs                                                                                                                                 |

## BRSR Core (top 150 listed companies)

|                    |                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | SEBI — mandatory from FY2024-25                                                                                                |
| Why it matters     | More stringent than standard BRSR. Mandatory assurance of ESG data. Important if any CarbonSense clients are listed companies. |
| Applies to         | Listed companies                                                                                                               |

## Pollution Control Board Consent to Operate

|                    |                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authority / Source | State PCBs                                                                                                                                                        |
| Why it matters     | Consent to Establish (CTE) and Consent to Operate (CTO) are legally required for any manufacturing unit. Renewal tracking is a compliance management opportunity. |
| Applies to         | All manufacturing SMEs                                                                                                                                            |

13. Final System Flow Summary

The end-to-end compliance system flow across all components:

|                        |                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Data Input             | CSV ingestion / manual entry / OCR receipts / evidence uploads feed into the emission calculation engine                                    |
| Emission Calculation   | GHG Protocol engine computes Scope 1, 2, 3 totals, category breakdown, and KPI snapshots                                                    |
| Policy Matching        | Org industry profile and emission footprint are matched against the policies table to identify applicable policies                          |
| Compliance Checks      | Three check types run: Data (metrics present and changed), Action (steps completed), Reporting (submissions made)                           |
| Verification Engine    | Rule engine validates each check — data change vs baseline, OCR extraction confirmation, report submission timestamp                        |
| LLM Assist (Groq)      | Explains requirements, suggests improvements, extracts metrics from OCR output — never sets verified status directly                        |
| Score Calculation      | $\text{data\_score (40)} + \text{action\_score (40)} + \text{reporting\_score (20)} = \text{total out of 100}$ , persisted to score history |
| Recommendations + TEME | Score gaps feed into the recommendation engine. TEME provides offset planning for residual emissions after reduction actions                |

Built with scientific rigor. No greenwashing. Reduction-first, always.

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